

Spectrasense

meeting 1 - meeting minutes

Discussion points:

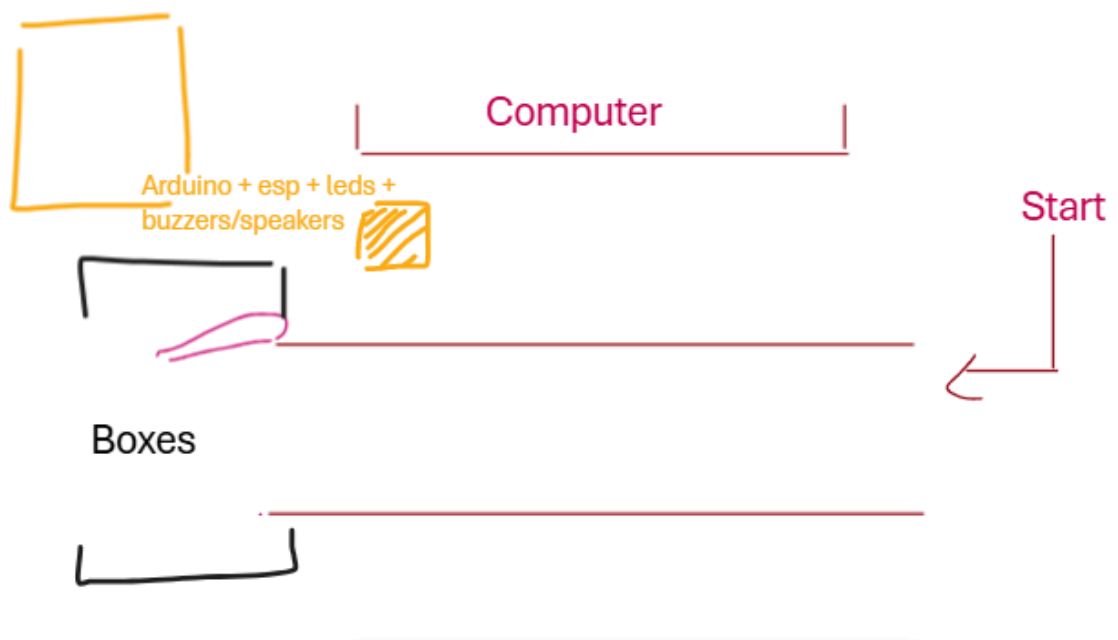
1) Project Flow [Important]:

Before the conveyor belt starts, the user is given two options:

1- category classification

2- metal classification

Item will be dropped in the start of the conveyor belt, then once it reaches in front of the computer, the conveyor belt stops, giving the project the space to decide what this item is



- Expected output:

Type: [silver/gold/metal/etc] [bottles/papers/pens]

Category: [key/accessory/tool/etc]

- The project should classify the object into categories using **computer vision**. And should classify the metal type using both **hardware** and **computer vision**
- If user is in mode 1 [category classification], the project should divide items into **three** boxes using servos
- If user is in mode 2 [metal classification], the project should divide items into **three** boxes using servos

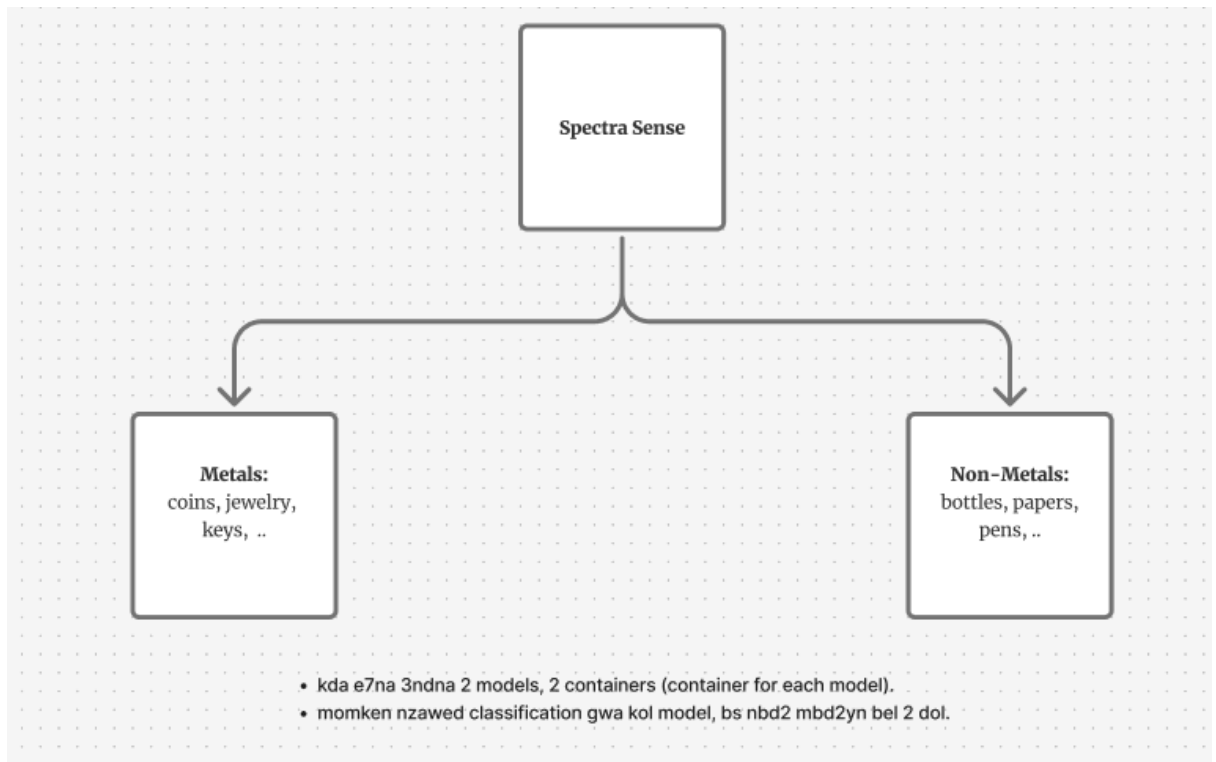
2) Project components and aspects discussed:

- hardware metal detection: either using a hand-made circuit, or using an inductive proximity sensor PNP (link: [product](#))
 Note: another sensor similar was discussed (NPN) but no conclusion was made.
- we discussed the options of using either a magnetic field sensor if this would advance our hardware classification
- **display**: we discussed using **LCD i2c/OLED/CYD esp32**
- **Conveyor belt**:
 We will make 1 conveyor belt using **2x 360 servo Motors**, and attach 3 small **servo** arms to move objects into the classification boxes like said before
 Inspo resource: [Inspo vid \[Important\]](#)
- if item placed in the start of the conveyor belt is not a metal and detected that it is not, the conveyor belt will move backwards to place that item at the start again [with display: not a metal]

Extra discussions:

We wanted to make the project interactive, so we will add LEDs[green and red], and high voltage buzzers/speakers for fun noises, etc.

We discussed if we can use laptop's camera for **computer vision**, or if we can use a ESP32 CAM: [product link](#)



IoT Team Leader: Omar Nabil

AI Team Leader: Mohamed Abdelnaser